CommentGuard AI · Interview Presentation Script & Guide

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SLIDE 1: Title, Problem Statement & Proposed Solution

Time: ~1.5 Minutes

Key Numbers: 20,000 comments/day | 2 human moderators | 94% automation | ~320 human review queue

Slide Purpose: Hook the evaluators by clearly framing the physical impossibility of manual review, then present your 3-tier cascade as the clean architectural solution.

Word-for-Word Speaking Script:
"Good morning evaluators. My name is Dhanush Bheemsetty, and today I am presenting CommentGuard AI—an intelligent moderation system designed for high-volume video platforms.

The problem statement presents an extreme constraint: A popular cooking video platform receives 20,000 user comments per day, filled with spam links, promotional ads, and abusive remarks. However, the platform has only 2 human moderators. In an 8-hour shift, two people can realistically review about 1,000 to 1,200 comments combined. Manually checking 20,000 comments is physically impossible.

My solution is a 3-Tier AI Cascade that chains deterministic rules, classical machine learning, and a lightweight deep learning transformer. It automatically handles 94% of comments with high precision, leaving only about 320 truly ambiguous cases for the two moderators (~160 each, taking under 45 minutes). Let me walk you through how this works."

SLIDE 2: The 3-Tier Cascade Architecture (AI ⊃ ML ⊃ DL)

Time: ~2 Minutes

Key Numbers: Tier 1 (~1.2ms) | Tier 2 (~4.8ms) | Tier 3 (~14.5ms) | Total Cascade ~20ms | Cost <$2/month

Slide Purpose: Demonstrate deep understanding of AI vs ML vs DL hierarchy. Explain why you picked a Small Transformer (DistilBERT) instead of an expensive, slow LLM.

Word-for-Word Speaking Script:
"On Slide 2, we see how the three AI tiers work together as a cascade. The design principle is simple: use the cheapest tool that is confident enough, and escalate only when needed.

Tier 1 is Deterministic AI Rules. Running in just 1.2 milliseconds at zero compute cost, it catches obvious bit.ly links, telegram handles, and known abusive words. It also includes our cooking allowlist to protect safe culinary verbs.

Tier 2 is Classical Machine Learning. Using TF-IDF word frequencies, uppercase shouting ratios, and Logistic Regression, it evaluates bulk comments in 4.8 milliseconds. For example, a comment with all caps like 'DM ME FOR CHEAP FOLLOWERS NOW' scores 94% spam.

Tier 3 is our Small Transformer—specifically DistilBERT with 66 million parameters. I intentionally chose a small transformer instead of a massive LLM because it runs in only 14 milliseconds on a normal CPU and costs less than $2 a month for 20,000 comments! Most importantly, it solves context: when a fan writes 'This risotto recipe is killer, I would die for this dish!', DistilBERT recognizes enthusiastic praise, NOT violence, preventing false flags.

Finally, Tier 4 is our two human moderators. They only receive the ~320 borderline cases where confidence is between 40% and 89%."

SLIDE 3: Core Questions (Part 1: Classification & Actions)

Time: ~1.5 Minutes

Key Numbers: Fine ~57% (11,400) | Spam ~21% (4,200) | Abusive ~5% (900) | Unclear ~18% (3,500 total, top 320 to humans)

Slide Purpose: Directly answer Question 1 and Question 2 from the problem PDF with concrete confidence thresholds and operational actions.

Word-for-Word Speaking Script:
"Slide 3 answers the first two questions from your problem statement.

Question 1: How do we classify comments? We compute calibrated probability distributions: P(fine), P(spam), and P(abusive). If confidence is 90% or higher, the AI acts automatically. If confidence falls between 40% and 89%, it is classified as 'Unclear'.

Question 2: What should happen to each category?
1. Fine comments (~57%, or 11,400 daily) are Auto-Published immediately with zero creator friction.
2. Spam comments (~21%, or 4,200 daily) are Auto-Hidden and logged. Repeat offenders are shadowbanned.
3. Abusive comments (~5%, or 900 daily) are Auto-Hidden and flagged for community policy review, issuing account strikes.
4. Unclear comments are sent to the Human Queue. The two moderators review the top ~320 priority cases, taking only 45 minutes."

SLIDE 4: Core Questions (Part 2: Precision & Continuous Learning)

Time: ~1.5 Minutes

Key Numbers: Asymmetric 90% threshold | ~300 daily labels | <5 min weekly retraining | 2.5% drift threshold

Slide Purpose: Answer Question 3 and Question 4, proving your system prevents false positives on cooking channels and dynamically adapts to evolving spam tricks.

Word-for-Word Speaking Script:
"Slide 4 answers the remaining two questions: avoiding false positives and staying ahead of new spam.

Question 3: How do we avoid hiding genuine comments? On cooking channels, words like 'kill the heat' or 'beat the eggs' look violent to naive keyword filters. DistilBERT solves this through context awareness. In addition, we require 90% confidence to auto-hide. If the AI is only 75% sure, it does NOT hide the comment—it routes to human review. Creators also get a 'Held for Review' panel to restore comments with 1 click.

Question 4: How do we keep up with new spam tricks? Spammers mutate tricks using spaced letters like 'c-h-e-a-p' or shortener links. We handle this through adversarial text normalization and an active learning feedback loop. Every decision made by our 2 moderators becomes a labeled training sample—generating ~300 fresh labels daily. Every week, DistilBERT is fine-tuned on these labels in under 5 minutes, ensuring our AI learns new tricks from real data."

SLIDE 5: Live Demo Link & Conclusion

Time: ~1 Minute

Key Numbers: Live URL https://commentguard-ai.onrender.com/ | Deployed on Render Global CDN

Slide Purpose: Direct evaluators to the live deployed URL, summarize key engineering achievements, and invite Q&A; and live testing.

Word-for-Word Speaking Script:
"To conclude, I didn't just write down the theory—I built, tested, and deployed the complete working prototype. It is live right now at https://commentguard-ai.onrender.com/.

In the prototype, evaluators can fetch real comments from YouTube cooking videos, type any raw comment to see the 3-tier cascade diagnose it live in ~20 milliseconds, and test the moderator decision queue and simulated retraining.

Thank you very much for your time. I am now open to your questions and would love to give you a live demonstration!"

Anticipated Evaluator Questions & High-Impact Answers:

Q: Why DistilBERT over GPT-4 or LLaMA?
A: Latency and cost: DistilBERT takes ~14ms on ordinary CPU instances and costs <$2/month for 20k comments, while GPT-4 takes ~1000ms and costs over $200/month. DistilBERT also fine-tunes in minutes on our moderator feedback loop.

Q: What if the 2 moderators fall behind?
A: The review queue is strictly prioritized by risk score. Furthermore, creators have a self-service 'Held for review' tool to approve comments on their own videos, acting as a crowd-sourced moderation extension.

Q: How do you measure model drift?
A: We track two core metrics weekly: False Positive Rate (benchmark <2.5%) and Spam Escape Rate (benchmark <1.0%). If the false positive rate spikes, automated threshold recalibration triggers.